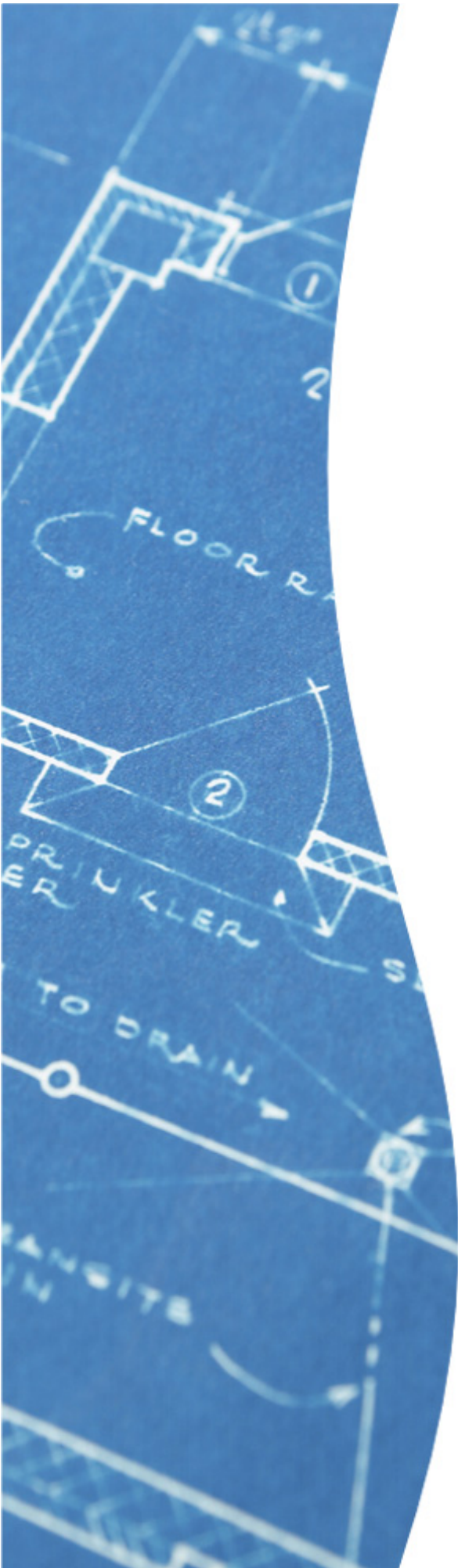


A vertical, curved architectural drawing on the left side of the page. It shows a floor plan with various rooms and corridors. Labels include "FLOOR R", "PRINKLER", "TO DRAIN", and "TRANSITS". There are also circled numbers "1" and "2" and a dimension line labeled "21' 0\"/>

USING SUN SYSTEMS TO BUILD A VIRTUAL AND DYNAMIC INFRASTRUCTURE

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Table of Contents

Using Sun Systems to Build a Virtual and Dynamic Infrastructure	1
The Dynamic Datacenter	1
Technology Under Evaluation	2
Sun™ Logical Domains	2
Scalent Virtual Operating Environment (V/OE)	3
Architecture Overview	8
Hardware and Software Requirements	9
Proof-of-Concept Architecture	10
Installation Overview	11
Controller Installation	11
Client Configuration Options	12
Disk-Booted Personas	13
Using Solaris ZFS to provide NFS and iSCSI storage	13
Installation for Network-Booted Linux Clients	15
Linux: NFS-Booted System Configuration	15
Linux: iSCSI-Booted System Configuration	16
Installation for Solaris OS Clients	19
Using Logical Domains (LDoms)	19
Installation of Diskless Solaris Environment	21
Installation for Windows Clients	27
Windows Disk-Booted Systems	27
Windows iSCSI Network-Booted Systems	27
Summary	30
About the Author	30
Acknowledgements	30
References	31
Ordering Sun Documents	31
Accessing Sun Documentation Online	31
Appendix A: Supported Ethernet Switches	32
Appendix B: Config_tftp Script	34
Appendix C: Scalent V/OE Pre-Installation Checklist	37
Appendix D: DHCP Manager Wizard	42
Appendix E: Scalent Controller Installation	45

Using Sun Systems to Build a Virtual and Dynamic Infrastructure

Today's rapidly changing business environments require a flexible infrastructure that can quickly and easily adapt to evolving needs. A dynamic datacenter infrastructure, using technologies such as Sun™ Logical Domains and the Scalent™ Virtual Operating Environment (V/OE), can provide a means to easily provision and repurpose servers upon demand.

This document describes a proof-of-concept exercise that deployed a virtual and dynamic datacenter infrastructure using the Sun Blade™ 6000 modular system, Logical Domains, and the Scalent V/OE software, with clients running Windows, Linux, and Solaris™ operating systems. Intended as a starting point in designing a dynamic datacenter architecture, this document addresses the following topics:

- “The Dynamic Datacenter” on page 1 describes the need for a dynamic datacenter.
- “Technology Under Evaluation” on page 2 provides an overview of the Logical Domains and Scalent V/OE technologies.
- “Installation Overview” on page 11 introduces the installation procedures used in this proof-of-concept exercise.
- “Installation for Network-Booted Linux Clients” on page 15, “Installation for Solaris OS Clients” on page 19, and “Installation for Windows Clients” on page 27 outline the procedures needed to install clients running the various operating systems.

The Dynamic Datacenter

In today's rapidly changing environments with variable application workloads, flexibility is essential. The ability to reprovision servers quickly and easily while providing sufficient infrastructure to support peak demands is an important part of this flexibility. The ability to shut down servers when they are not needed, to save resources including power and cooling, is equally important as businesses strive to reduce operating costs.

A server that is sitting idle or doing minimal work has a very low work to power consumption ratio. As the server produces more work, the ratio increases significantly—a server with high utilization is more power efficient than an idle one. In other words, the more highly utilized the server, the less energy is wasted by that server.

Enterprises can leverage virtualization technologies to consolidate multiple applications or server workloads onto a single physical machine. This approach can help a business provide all the necessary services, while avoiding the expense of multiple idle (or mostly idle) servers that are simply heating the datacenter floor. A

variety of virtualization technologies are available today, including Logical Domains that use the on-board hypervisor of Sun UltraSPARC T1/T2/T2 Plus-based servers with Chip Multithreading (CMT) technology; and VMware, Hyper-V and Xen implementations on x86 servers.

Virtualization is a flexible and powerful solution that can be used to consolidate services, reduce management costs and improve utilization. However, many enterprises are not able to fully virtualize datacenters in their entirety at one time. Instead, they are adopting these technologies over time where it makes sense to them. As they become comfortable using the virtualization technologies they have chosen, they can broaden the use of them.

The Scalent Virtual Operating Environment (V/OE) allows both the physical environment and the virtualized environment to coexist, leveraging the virtues of both worlds and even allowing them to overlap. Using V/OE software, customers can deploy systems in a physical environment and potentially move to a virtual environment, or vice-versa. By combining the Scalent V/OE with virtualization technologies such as Logical Domains, an enterprise can deploy a flexible datacenter architecture with the ability to provision services dynamically on both physical or virtual machines.

Technology Under Evaluation

The following sections provide an overview of Sun Logical Domains and the Scalent Virtual Operating Environment (V/OE) software, two virtualization technologies employed in this proof-of-concept exercise.

Sun™ Logical Domains

Sun Logical Domains (LDoms) technology, supported on all Sun servers which utilize Sun processors with Chip Multithreading Technology, enable a system's hardware resources to be subdivided creating partitions called *logical domains* (Figure 1). Each logical domain is a full virtual machine that runs an independent operating system instance and contains virtualized CPU, memory, storage, console, and cryptographic devices.

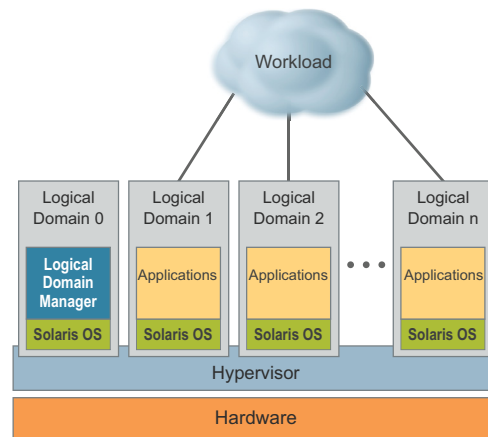


Figure 1. Sun's Logical Domains architecture.

A *control domain* (Logical Domain 0) runs Logical Domain Manager software; this software is used to create and manage additional logical domains called *guest domains*. A thin layer of firmware called the *hypervisor*, resident on Sun UltraSPARC® T1, T2, and T2 Plus processors, enforces resource management at the hardware level. The hypervisor provides a stable, virtualized machine architecture to which an operating system can be written. As such, each logical domain is completely isolated and the maximum number of virtual machines created on a single platform relies upon the capabilities of the hypervisor as opposed to the number of physical hardware devices installed in the system.

By taking advantage of Logical Domains, organizations gain the flexibility to deploy multiple operating systems simultaneously on a single platform. In addition, administrators can leverage virtual device capabilities to transport an entire software stack hosted on a logical domain from one physical machine to another. Logical Domains can also host Solaris Containers to capture the isolation, flexibility, and manageability features of both technologies. By deeply integrating logical domains with both the industry-leading chip multithreading (CMT) capability of the Sun UltraSPARC T1, T2 and T2+ processor and the Solaris 10 OS, logical domains technology increases flexibility, isolates workload processing, and improves the potential for maximum server utilization.

Note – For more detail about LDomS, see *Beginners Guide to LDomS: Understanding and Deploying Logical Domains for Logical Domains 1.0 Release*, available on the Web at <http://www.sun.com/blueprints/0207/820-0832.html>.

Scalent Virtual Operating Environment (V/OE)

The Scalent V/OE software enables administrators to manage physical and virtual servers in a datacenter, and repurpose those machines without requiring changes to the physical machine, cables, LAN or SAN access.

The Scalent V/OE software includes three primary components: the *Controller*, the *Agent Software*, and the *Console*. A *Software Development Toolkit (SDK)* is also available.

- *Controller*

The Scalent Controller manages the physical and virtual hardware, software, and network configurations in the managed environment. Each configuration requires one Controller; a second Controller can be configured to act as an optional hot standby.

- *Agent Software*

The Scalent Agent software, installed on every managed server in the configuration, obtains configuration information and provides status information to the Controller. The Scalent software maintains a heartbeat between the Controller and Agent Software; if the heartbeat fails, the Controller selects another physical or virtual machine to run that persona.

- *Console*

The Scalent Console provides a graphical interface to the Controller, and is used to configure and monitor the components (both physical and virtual) in the managed environment.

- *Scalent Software Development Toolkit (SDK)*

The Scalent SDK provides a command-line interface, plus Java and Web services interfaces that can be used to automate tasks performed by the Scalent Console.

Scalent Controller Management Interface

The Scalent Controller Management Interface provides multiple views of the physical and virtual components in the V/OE environment. Figure 2, depicting a physical view of the Scalent V/OE managed environment, offers an overview of the servers that are available to the Scalent Controller. This view provides a wealth of information, including:

- *All the servers discovered by the controller.* Note some servers have small boxes within, which represent a virtualized environment; each internal box represents a virtual server.
- *Which personas are running on the physical or virtual servers.* A persona is the definition of a server environment captured on disk, including the operating system, Scalent agent software, application software, and the network and other settings required to run an application on a server in the Scalent environment.
- *What operating system is booted and which application is running.* Operating system icons are automatically assigned by the Scalent software, and logos or other identifiers for applications can be assigned by the administrator, helping to easily identify what application or function the server is providing.

- *Network or disk-booted status.* A lightning bolt symbol indicates that the server (such as a Solaris physical server or a Solaris Logical Domain) is network booted. A hard drive symbol indicates the system (such as the ESX or LDom servers) is disk booted.
- *Persona assignment.* The lock symbol indicates that the persona assigned to that server can only boot on that server. Personas that are network booted can be assigned to any available server unless they are bound to a specific server this way.
- *Port connections and status.* The number to the right of the server indicates which port the server is connected to on the managed switch. If the light is green, the port is active, if it is off or red, the port is inactive.



Figure 2. Physical view of the Scalent V/OE managed environment.

If a server is selected, then the relevant information for this server is displayed on the right. Options to explore even more server-specific data is available with the drop down menu on the top right. The data on the right pane can be edited and saved which modifies the configuration of the server's description or the persona assigned to it. Colors indicate the state of the servers:

- Gray – the server is off
- Red – the server is being powered on or off, or the persona is down
- Yellow – the persona is booting or shutting down
- Green – the server is powered on and the persona is running

The catalog view, shown in Figure 3, can display servers that have been discovered, all personas that have been defined, or all the virtual racks that are available in the environment.

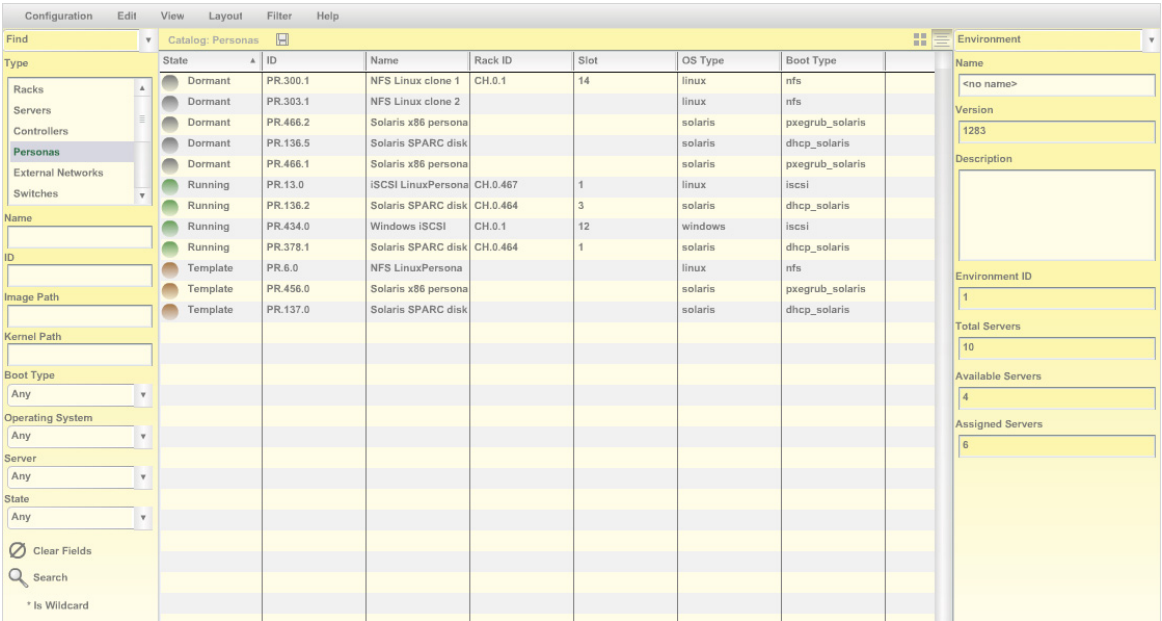


Figure 3. Catalog view of the Scalent V/OE managed environment.

Figure 4 depicts a virtual view of the Scalent V/OE managed environment, showing the state of the *library* of personas available in the controller. If a persona is highlighted and the view is changed to *Physical view* (as shown in Figure 2), the server assigned to that persona is highlighted. While not explicitly represented on these diagrams, all servers are connected to the boot network by default.

This virtual view offers a more functional perspective of what is available in the managed environment, and allows different personas to be assigned virtual NICs and networked together using the clouds and vSwitches on the left. The *cloud* represents connectivity to an infrastructure element that is outside the control of Scalent — for example, an external network that could be another server, switch, router, load balancer, firewall or any other network device. The *vSwitch* is a representation of a 802.1q-based VLAN that gets created on one of the physical network switches that is part of the Scalent environment.

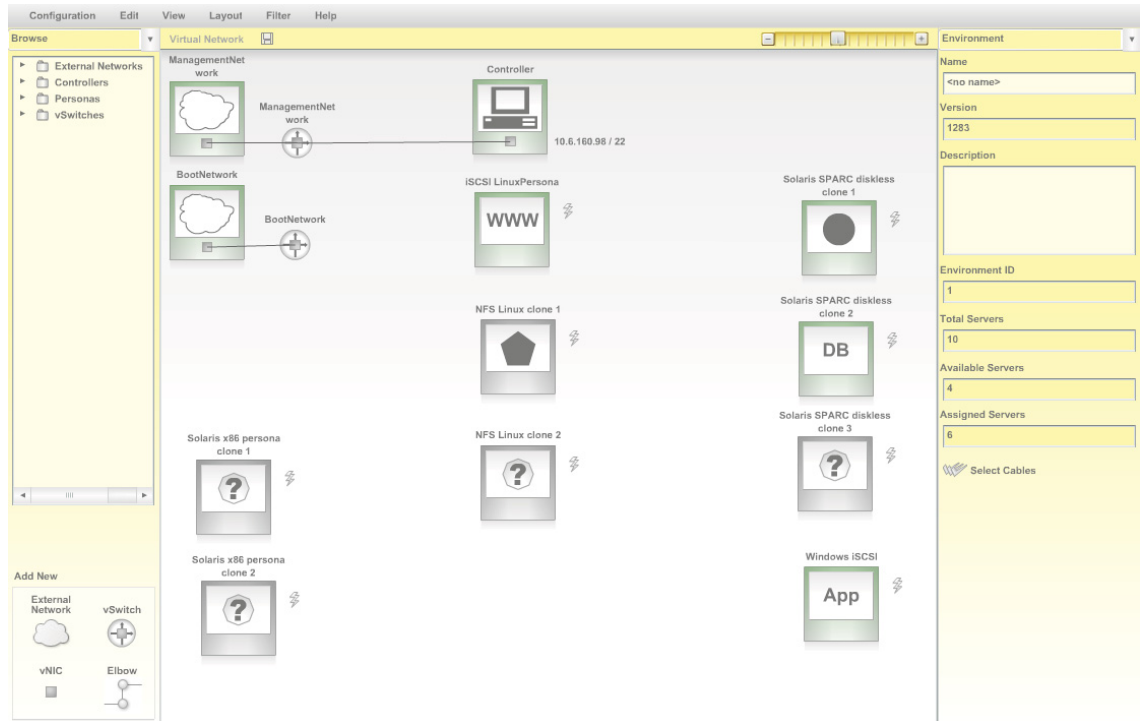


Figure 4. Virtual view of the Scalent V/OE managed environment.

Architecture Overview

The Scalent environment requires two networks — a boot network and a management network — for the basic architecture, as shown in Figure 5. In the architecture employed by this proof-of-concept scenario, a server was designated to run the Scalent Controller software and a second server was utilized as a file server. The remaining servers ran the Scalent Agent software, and were managed by the Scalent environment.

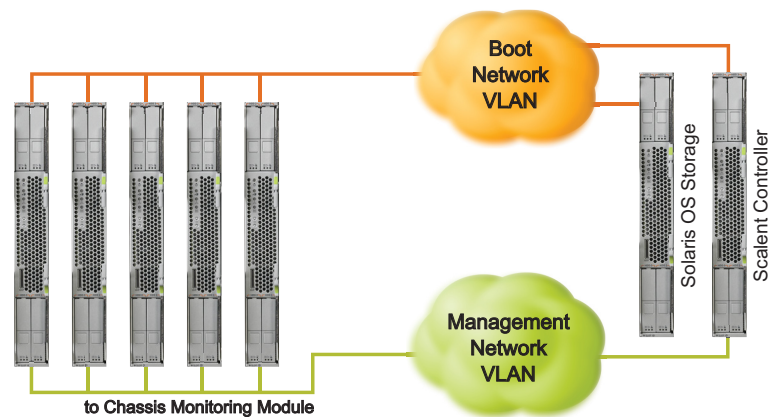


Figure 5. Basic architecture.

- *Boot Network* – The boot network provides the servers with a means to be discovered by the Controller. This network also provides servers with their identity, or *persona* (as it is called in the Scalent environment), allowing the server to be network booted using NFS or iSCSI. The NFS and iSCSI storage servers also require access to this network.
- *Management Network* – The management network allows communication between the Scalent Controller and the Service Processors of the servers in order to control power to the servers. This connectivity allows more automation of deployment of services, as the Controller can select individual servers for specific tasks.

In addition to these two networks, other networks may be necessary in order to provide a data, service or intercommunication network to the servers. Any additional networks can be managed by the Scalent Controller or managed independently. However, the Controller offers a convenient interface with which these networks can be created virtually, leveraging the VLAN capability of the switches and the dynamic management interface of the Controller.

Hardware and Software Requirements

The Scalent V/OE software is supported on an extensive range of hardware, including servers, switches, storage, and Host Bus Adapters (HBAs). In addition, Solaris OS, Linux, Windows, and VMware are all supported on the managed servers. Choosing supported components is important to a successful deployment of the Scalent V/OE environment, as this allows the Scalent Controller to communicate successfully with all systems in the architecture.

Supported Hardware

A variety of x86 and UltraSPARC CMT rack-mount and Sun Blade server modules are supported, providing flexibility for deployment scenarios. The supported servers include:

- Sun Blade 6000 Modular System
 - Sun Blade X6250, X6450, and X6220 server modules
 - Sun Blade T6300 and T6320 server modules
- x86 rack mount servers:
 - Sun Fire X4150, X4250, and X4450 servers
 - Sun Fire X4140, X4240, and X4440 servers
- UltraSPARC CMT rack mount servers:
 - Sun Fire T1000, Sun Fire T2000
 - Sun SPARC Enterprise T1000, Sun SPARC Enterprise T2000
 - Sun SPARC Enterprise T5120, Sun SPARC Enterprise T5220
 - Sun SPARC Enterprise T5140, Sun SPARC Enterprise T5240
 - Netra T5440
 - Sun SPARC Enterprise T5440
- UltraSPARC CMT Logical Domains

The following hardware requirements apply to Ethernet switches, Host Bus Adapters (HBAs), and Fibre Channel Storage Array Network (SAN) switches:

- *Ethernet switches* — A broad range of Ethernet switches, including Cisco and Foundry products, are supported. See Appendix A, “Supported Ethernet Switches” on page 32 for a listing of switches supported at the time of publication.
- *HBAs* — QLogic and Emulex HBAs, two of the most widely-used adapters for SAN connectivity, are supported. These HBAs are available for all supported servers, including rack-mounted servers and Sun Blade server modules.
- *Fibre Channel SAN switches* — There are no specific requirements on the Fibre Channel SAN switches, because these switches are not managed by the Scalent Controller. Rather, these switches require user management to make the necessary volumes available to the servers.

Operating System Support

The server selected to run as the Scalent Controller must support Red Hat Linux 4 Update 4 32-bit. The operating environments supported for the client or virtual machines at the time of writing are:

- Solaris 10 OS — both x86 and SPARC
- Linux — Red Hat Linux 4 and 5 and Novell SLES 10
- Windows — 2003 Server and Enterprise
- VMware 3.0 and later

Proof-of-Concept Architecture

This proof-of-concept exercise used a Sun Blade 6000 Modular System, populated with seven Sun Blade server modules, as the basis for the hardware architecture. The seven server modules included:

- Three Sun Blade X6250 Server Modules
- One Sun Blade T6320 Server Module
- One Sun Blade X6450 Server Module
- Two Sun Blade X6220 Server Modules

A Sun Blade X6250 server module was used as the Scalent Controller. A second Sun Blade X6250 server module was used as a file server and for iSCSI target storage. Logical Domains were configured on the Sun Blade T6320 server module. This Sun Blade T6320 server module and the remaining server modules were configured with the Scalent Agent software and were managed in the Scalent V/OE environment.

These server modules were chosen to provide a variety of client platforms. The Sun Blade X6450 server module is diskless, making it an ideal candidate for a network-booted client. Once operating system images are available on the Controller, these diskless systems can easily be booted over NFS or iSCSI.

Note – The Sun Blade X6250 server module used in this exercise as the Scalent controller did not run the supported version of the Red Hat Linux operating system, but instead used the RedHat 5 Update 6 32-bit version. While using a non-supported version of the operating system is not recommended, it sufficed for proof-of-concept purposes. The latest release of Scalent V/OE is now supported to run on RedHat Enterprise Linux 5 Update 1.

Installation Overview

The basic Scalent V/OE deployment requires the Scalent Controller software to be installed on a server in the network. The Controller initializes the first switch it is connected to, and sets up VLANs for the boot and management networks.

After the Controller is set up, the systems that will be under the control of the V/OE can be discovered. All this requires is to connect a network interface to the boot network, connect the Service Processor network interface to the management network, and then reboot the server.

The Controller provides a PXE boot image that recognizes or identifies the server's capabilities, such as memory, number of processors and cores, hard drives, and whether it's configured to boot from hard drives or the network.

Once the hardware recognition is done, the systems can be booted normally and the Scalent Persona (or agent) software installed. The persona communicates with the Controller to obtain configuration information and provide status on the health of the server. If communication is lost with the Controller and the managed server, the Controller attempts to restart the persona elsewhere to provide continuity of service.

Once the system has been identified, it is added to the Controller's catalog and it is ready to be used. The systems are represented individually on the graphical console, and are organized based on the capabilities and the environment they are running.

Controller Installation

The Scalent Controller software runs on RedHat 4 update 4, and is installed using the Scalent Controller installation script. Basic network and system configuration information should be collected before running the starting the installation. A checklist is provided in the Scalent documentation which includes all information that is required. This information is summarized in Appendix C, "Scalent V/OE Pre-Installation Checklist" on page 37. Many default parameters are appropriate for most installations. While all parameters can be adjusted, the default values are recommended unless they need to be adjusted for the particular deployment environment.

In this example proof-of-concept exercise, default values were used except for those pertaining to the management network. These modifications were required, as the boot network represents a new network that has not been used before in the existing infrastructure, yet the servers have already been assigned management network information. This allows the Controller to easily be installed in existing datacenter environments with minimum disruption.

Before running the script, adequate connections should be established between the server and the switch being used. Figure 6 shows the physical connections used in this example configuration.

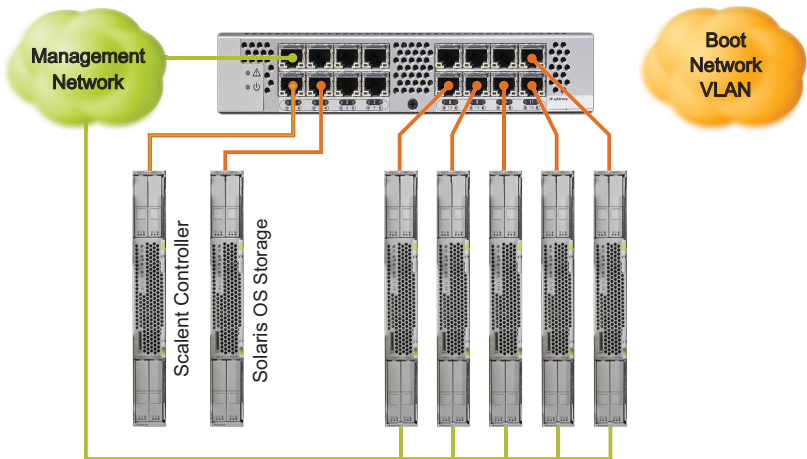


Figure 6. Physical switch connections.

Next, run the Scalent Controller installation script, which installs the Controller software and configures the switch. To run the script, mount or insert the CD that contains the Scalent software, go to the mounted directory on the server, copy the installation files to a directory on the hard drive, and run the installation script. For example:

```
# cd /media/cdrom/
# cp -rp ./installation /root
# cd /root/installation
# ./setup.sh
```

The output of a sample run of the Controller installation, including the inputs used for this sample configuration, is found in Appendix E, “Scalent Controller Installation” on page 45.

Client Configuration Options

Scalent V/OE clients can be booted from local disks, or across a network using iSCSI, NFS or a SAN. However, not all clients can be booted using all available methods.

Table 1 indicates which options are available for each type of operating system:

Table 1. Supported boot methods for Scalent V/OE clients.

OS Type	Disk	Boot Methods		
		iSCSI	NFS	SAN
Solaris x86	Yes	No	Yes	Yes
Solaris SPARC	Yes	No	Yes	Yes
Linux	Yes	Yes	Yes	Yes
Windows	Yes	Yes	No	Yes
VMware	Yes	No	No	Yes

Disk-Booted Personas

Clients that boot operating systems from the hard drive are supported as disk-booted personas in the Scalent V/OE environment. These disk-booted systems, or personas, simply need to have the operating system software and the Scalent Persona (agent) software installed locally. When the system reboots, the physical server and its corresponding persona appear in the Scalent Controller graphical console.

Each operating system has its own way of installing the persona agent software, but it essentially consists of three steps:

1. Install the operating system and all drivers on the local server.
2. Configure the network interface of the server to be on the boot network so that it can communicate with the Controller; this can use a static configuration or DHCP.
3. Run the setup script provided in the Scalent software distribution. This includes copying the 'installation' directory to a local file system on the server and then running the following command:
 - a. For Linux and Solaris systems, run `setup.sh` and select the '*P*ersona' option
 - b. For Windows servers, run the `setup_persona.exe` executable.

After the system reboots, the persona-enabled server will communicate with the Controller and the server will become available and visible on the Scalent V/OE interface.

Note – For Solaris diskless clients, see “Installation of Diskless Solaris Environment” on page 21.

Using Solaris ZFS to provide NFS and iSCSI storage

One way to simplify storage in a Scalent V/OE deployment is to use Solaris ZFS, which provides direct NFS and iSCSI target management and simple configuration options. If a server with enough storage is connected to the boot network, it can become the storage device for all network booted servers.

In order to configure ZFS to offer NFS volumes and iSCSI targets, follow these configuration steps:

1. Create a ZFS storage pool using available devices in the server. The following command uses disk devices `c1t0d0`, `c1t1d0`, and `c1t2d0` to create a pool named `shares`:

```
# zpool create shares c1t0d0 c1t1d0 c1t2d0
```

2. Create a filesystem on the pool and share it via NFS:

```
# zfs create shares/nfs
# zfs set sharenfs=rw,anon=0 shares/nfs
```

3. Create a volume to share using iSCSI (one per iSCSI-booted server). The following command creates and shares a 10 GB volume named `shares/iscsi-server1`:

```
# zfs -V 10g shares/iscsi-server1
# zfs set shareiscsi=on shares/iscsi-server1
```

4. On systems with controllers that have cache, NFS/ZFS slows down with certain write operations. A change to the ZFS kernel parameters alleviates this issue. The server used in this proof-of-concept exercise did have NVRAM, and found the following change beneficial when the NFS copy scripts indicated below were executed:
 - a. To make the change on the running system:

```
# echo zfs_nocacheflush/w0t1 | mdb -kw
```

- b. To make the change permanent, add the following line to `/etc/system`:

```
set zfs:zfs_nocacheflush=1
```

Note – In order to determine if modifications are required, visit the *ZFS Evil Tuning Guide* at http://www.solarisinternals.com/wiki/index.php/ZFS_Evil_Tuning_Guide and visit the *Cache Flushes* section.

Installation for Network-Booted Linux Clients

This section describes the procedures for installing network-booted clients running the Linux operating system. The general procedure involves first installing the operating system on a disk, followed by installing the Scalent Persona software on the client.

In Linux environments, preparations are required for both NFS and iSCSI booted personas. Specifically, an initial RAM disk (`initrd` file) and boot kernel must be generated and copied to the Controller so that it may boot the different kernel versions that the personas may need. In addition, iSCSI deployments may require certain parameters to be configured to allow for successful booting of the disk image.

If both iSCSI and NFS boot methods are planned, create the iSCSI-enabled RAM disk first, as described in the section “Linux: iSCSI-Booted System Configuration” on page 16. This can then be used for both NFS- and iSCSI-booted servers.

Linux: NFS-Booted System Configuration

NFS-booted personas require an NFS server available in the boot network. The server that is going to be used as the original for the NFS-booted persona should be able to access this NFS storage server as well.

To create the NFS-booted persona, perform the following steps.

1. On the system to be replicated, install necessary software:
 - a. Install the supported Linux distribution that will be used.
 - b. Run the setup script provided in the Scalent software distribution to install the Scalent Agent software. To do this, copy the 'installation' directory from the mounted CD to a local file system on the server and then run the `setup.sh` command and select the '*P*ersona' option.
 - c. Install any other software that is expected to run on this server.
2. On this same system to be replicated, create a RAM disk for the Linux distribution:

```
# /opt/scalent/bin/mkscalentrd
```

This utility will indicate the location where the resulting RAM disk was written.

3. Copy that RAM disk and the kernel in the Linux distribution to the Controller directory as indicated below, substituting the correct Controller IP address. This RAM disk and kernel will be used for booting the servers using PXE/BOOTP capabilities.

```
# scp /tmp/initrd-2.6.9-34.ELsmp.img.gz
  controller_ip:/var/opt/scalent/tftpboot/ramdisk
# scp /boot/vmlinuz-2.6.9-34.ELsmp
  controller_ip:/var/opt/scalent/tftpboot/kernel
```

4. Once the system to be replicated is running, a single command generates the image on the NFS storage device, and then the personas are configured in the Controller using the Scalent SDK. Run the following commands to enable the configuration:

- a. On the system to be replicated, invoke the following command substituting the NFS server IP address, path, and persona name:

```
# /opt/scalent/bin/makenb.sh nfs_server_IP:path/Persona_name
```

- b. On the Scalent Controller console, use the SDK to add the persona, substituting the IP address of the NFS server, path and persona name:

```
# /opt/scalent/bin/sdk
>> open
>> add persona bootType=pxe_linux osFamily=linux osArch= x86_32
image=IPAddress:/Path/Persona_name kernel="vmlinuz-2.6.9-34.ELsmp"
>> save
>> close
```

5. The NFS-booted persona will now appear in the 'Virtual View' of the Controller's graphical console. This persona can now be booted on any network-bootable server that is available.

Linux: iSCSI-Booted System Configuration

Installing an iSCSI-booted persona requires an iSCSI storage device available in the boot network. The server that is going to be used as the original for the iSCSI-booted persona should be able to access this iSCSI storage device as well.

1. On the system to be replicated, install necessary software:
 - a. Install the supported Linux distribution that will be used.
 - b. Run the setup script provided in the Scalent software distribution to install the Scalent Agent software. To do this, copy the 'installation' directory from the mounted CD to a local file system on the server and then run the `setup.sh` command and select the '*P*ersona' option.
 - c. Install any other software that is expected to run on this server.

Note – The following instructions are specific for the Red Hat Linux 4 distribution. Other distributions, such as SuSE and Red Hat Linux 5, have slightly different instructions and are not covered in this document. See the Scalent and specific operating system distribution documentation for more details.

2. On the system to be replicated, confirm the iSCSI packages are installed on the Linux distribution being used. The systems in this example configuration installed the complete distribution of Red Hat Linux 4, and therefore had the required iSCSI package. To verify the availability of the needed packages in the current installation, run the following command:

```
# rpm -q -a | grep iscsi
```

This command should return the existence of a package named `iscsi-initiator-utils`.

3. On the system to be replicated, export the variable `sc_iscsird=yes` on the current environment, and then create a RAM disk for the Linux distribution:

```
# export sc_iscsird=yes
# /opt/scalent/bin/mkscalentrd
```

This utility will indicate the location where the resulting RAM disk was written.

4. Copy that RAM disk and the kernel in the Linux distribution to the Controller directory as indicated below, substituting the controller IP address. This RAM disk and kernel will be used for booting the servers using PXE/BOOTP capabilities with the necessary iSCSI modules to boot the image stored on the iSCSI storage device.

```
# scp /tmp/initrd-2.6.9-34.ELsmp.img.gz
  controller_ip:/var/opt/scalent/tftpboot/ramdisk
# scp /boot/vmlinuz-2.6.9-34.ELsmp
  controller_ip:/var/opt/scalent/tftpboot/kernel
```

5. Next, mount the iSCSI volume where the Linux image will reside. In order to do this, the server must be configured as an iSCSI initiator. Different Linux distributions do this differently and they are documented in the Scalent documentation. For the Red Hat Linux 4 distribution used in this example, the file `/etc/initiatorname.iscsi` must be edited to include the following line:

```
InitiatorName=initiator
```

Where *initiator* includes the iSCSI target name, but with a different ending to identify the initiator from the target. The following is an example of a target initiator pair:

```
target=iqn.2003-06.com.example:linux-rh4.target
initiator=iqn.2003-06.com.example:linux-rh4.initiator
```

Not all storage devices enforce target/initiator naming, but this step is important for storage devices that do. The iSCSI storage target discussed later does not enforce strict naming requirements.

The Red Hat Linux 4 distributions also requires configuration of the discovery IP address, which is the address of the storage device as configured on the network. This IP address should be configured on the file `/etc/iscsi.conf` by replacing the following value:

```
DiscoveryAddress = IP_address_of_iSCSI_storage
```

6. Make sure iSCSI initiator services and the configuration are current by restarting them using the following command:

```
# /etc/init.d/iscsi restart
```

7. Use the `iscsi-ls` command to verify that the iSCSI volume is now be available to the server:

```
# iscsi-ls -l |grep Device
Device: /dev/sdd
```

8. Next, copy the installed server to the iSCSI volume. To do this, run the following command on the system to be replicated:

```
# /opt/scalent/bin/copy_persona.sh -d /dev/sdd -r /dev/sda -t 1000 -n
```

This command copies the image to `/dev/sdd`, configures the image to later boot as `/dev/sda`, and provides the image with 1 GB of swap space. The `-n` is used to indicate this is a network booted system.

9. Lastly, configure the persona on the Scalent Controller using the SDK:

```
# /opt/scalent/bin/sdk
>> open
>> add persona bootType=iscsi osFamily=linux osArch=x86_32
image="iscsi|netapp|172.18.1.100|3260|iqn.2003-06.com.example:linux-
rh4.target|iqn.2003-06.com.example:linux-
rh4.initiator|/dev/sda1|ext3|iscsi" kernel="vmlinuz-2.6.9-34.ELsmp"
name="iSCSI Linux RH4 Persona"
>> save
>> close
```

The iSCSI booted persona will appear in the graphical console of the Controller. This persona can now be booted on any network bootable server that is available.

Installation for Solaris OS Clients

Environments that include Sun CMT servers (with UltraSPARC T1, T2, or T2 Plus processors) have the option of configuring Logical Domains on those servers. With Logical Domains, a system's hardware resources can be allocated into separate logical domains, full virtual machines each running an independent instance of the operating system.

The following sections describe how to configure Logical Domains (an optional step), and how to configure the Scalent V/OE environment to support diskless Solaris OS clients.

Using Logical Domains (LDomS)

Logical Domains must first be created before the Scalent Controller can recognize them. The following sections describe how to create a primary domain and one or more guest domains. For further information on Logical Domains please visit

<http://www.sun.com/ldoms>.

Primary Domain Configuration

The following steps are used to create the primary, or control, domain on the system. The primary domain must be created before any additional guest domains are configured:

1. Install the system with Solaris 10 update 5 fully patched or Solaris 10 update 6.
2. Next, install and enable the Logical Domains 1.0.3 software package:

```
# pkgadd -d <path to package>/SUNWldm
# PATH=/opt/SUNWldm/bin:$PATH ; export PATH
# svcadm enable ldmd
```

3. Configure the primary (control) domain:

- a. Eliminate the use of a MAU in the primary domain:

```
# ldm set-mau 0 primary
```

- b. Assign CPUs (four, in this example) to the primary domain:

```
# ldm set-vcpu 4 primary
```

- c. Assign memory (4 GB in this example) to the primary domain:

```
# ldm set-mem 4G primary
```

- d. Add a virtual disk server:

```
# ldm add-vdiskserver primary-vds primary
```

- e. Add a virtual switch, using physical network device e1000g0:

```
# ldm add-vswitch net-dev=e1000g0 primary-vsw0 primary
```

- f. Add a virtual terminal concentrator:

```
# ldm add-vconscon port-range=5000-5100 primary-vcc primary
```

- g. Assign a configuration name in the Service Processor:

```
# ldm add-spconfig ldm-config
```

4. Change the /etc/hostname.e1000g0 filename to /etc/hostname.vsw0:

```
# mv /etc/hostname.e1000g0 /etc/hostname.vsw0
```

5. Reboot the server to apply the changes:

```
# init 6
```

Guest Domains Configuration

Since the domains can be network booted, the minimum requirement for these domains includes a network interface attached to the virtual switch created in the primary domain, plus CPU and memory resources.

Create one or more domains, keeping in mind the availability of server resources. The following steps illustrate creating one guest domain named domain1.

1. Create the logical domain:

```
# ldm create domain1
```

2. Assign a number of CPUs to this logical domain:

```
# ldm add-vcpu number_virtual_CPUs domain1
```

3. Assign memory to this logical domain:

```
# ldm set-memory amount_of_memory domain1
```

4. Add a virtual network to this logical domain:

```
# ldm add-vnet vnet0 primary-vsw0 domain1
```

5. Specify the boot device:

```
# ldm set-variable boot-device=net:dhcp domain1
```

6. Bind resources to the newly-created logical domain:

```
# ldm bind-domain domain1
```

7. At this point the logical domain domain1 is ready to be booted, but it will look for a network boot server. If a boot server is already present in the environment, then simply start the domain using the following command:

```
# ldm start-domain domain1
```

If a boot server is not already present, proceed with the steps in the following section to set up these network services.

Installation of Diskless Solaris Environment

Before Solaris clients can be set up as network-booted clients in the Scalent environment, the diskless server configuration must first be created. Then, the Solaris network-booted client persona can be added to the Scalent Controller.

At a high level, this process involves the following steps:

1. Copy the Solaris distribution to the hard drive from CDs, DVDs or ISO images.
2. Configure diskless services. This can be done on any server, but in this proof-of-concept exercise the same NFS server was used.
3. Add and configure the x86 and/or SPARC diskless clients. For x86 diskless clients, a RAM disk must be created and DHCP must be configured before the server can be booted for the first time.

Installing Solaris Diskless Services

The Solaris OS diskless services need to be copied to the hard drive. In this proof-of-concept exercise, a separate Sun Blade X6250 server module was used as the NFS server for these Solaris OS files.

1. On the Solaris-based NFS server, download the Solaris 10 for SPARC and Solaris 10 for x86 media (DVD .iso files).
2. Mount the x86 .iso and go to the `media/Solaris_10/Tools` directory.
3. Copy CD / DVD distribution to a location on the hard drive such as:

```
bash-3.2# ./setup_install_server /shares/nfs/Solaris_10u5_i386
```

4. Switch the media for the SPARC DVD and repeat the process:

```
bash-3.2# ./setup_install_server /shares/nfs/Solaris_10u5_sparc
```

Configure Solaris Diskless Services

Continue by adding diskless services to the system. This will create the necessary directories and file structures needed by the Scalent environment in order to create the initial Solaris images. This process must be repeated for both x86 and each SPARC architecture. In this example, the sun4v and x86 architectures are demonstrated.

1. First, build the Solaris diskless services for the x86 platforms using the `smosservice` command:

```
bash-3.2# /usr/sadm/bin/smosservice add -- -x
mediapath=/shares/nfs/Solaris_10u5_i386/ -x
platform=i386.i86pc.Solaris_10 -x cluster=SUNWCXall
Authenticating as user: root

Type /? for help, pressing <enter> accepts the default denoted by [ ]
Please enter a string value for: password ::
Loading Tool: com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server
Login to boot-server as user root was successful.
Download of com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server was successful
```

2. Next, repeat the step for the SPARC media distribution:

```
bash-3.2# /usr/sadm/bin/smosservice add -- -x
mediapath=/shares/nfs/Solaris_10u5_sparc/ -x
platform=sparc.sun4v.Solaris_10 -x cluster=SUNWCXall
Authenticating as user: root

Type /? for help, pressing <enter> accepts the default denoted by [ ]
Please enter a string value for: password ::
Loading Tool: com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server
Login to boot-server as user root was successful.
Download of com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server was successful.
```

3. Confirm the process was successful and that the services are in place for both architectures:

```
bash-3.2# /usr/sadm/bin/smosservice list
Authenticating as user: root

Type /? for help, pressing <enter> accepts the default denoted by [ ]
Please enter a string value for: password ::
Loading Tool: com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server
Login to boot-server as user root was successful.
Download of com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server was successful.
Platform
-----
sparc.sun4v.Solaris_10
i386.i86pc.Solaris_10
```


Installing Solaris Diskless Clients

Next, add a sample diskless client for each architecture.

1. Use the following command to add an x86 diskless client:

```
bash-3.2# /usr/sadm/bin/smdiskless add -- -n diskless-x86
-i 172.18.1.65 -e 00:14:4f:fa:3a:12 -x os=i386.i86pc.Solaris_10
-x root=/export/root/diskless-x86 -x swap=/export/swap/diskless-x86
-x swapsize=512
Authenticating as user: root

Type /? for help, pressing <enter> accepts the default denoted by [ ]
Please enter a string value for: password ::
Loading Tool: com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server
Login to boot-server as user root was successful.
Download of com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server was successful.
```

2. Use the following command to add a SPARC diskless client:

```
bash-3.2# /usr/sadm/bin/smdiskless add -- -n diskless-sparc
-i 172.18.1.66 -e 00:14:4f:fa:3a:14 -x os=sparc.sun4v.Solaris_10
-x root=/export/root/diskless-sparc
-x swap=/export/swap/diskless-sparc -x swapsize=512
Authenticating as user: root

Type /? for help, pressing <enter> accepts the default denoted by [ ]
Please enter a string value for: password ::
Loading Tool: com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server
Login to boot-server as user root was successful.
Download of com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server was successful.
```

3. Verify that the diskless clients were added successfully:

```
bash-3.2# /usr/sadm/bin/smdiskless list
Authenticating as user: root

Type /? for help, pressing <enter> accepts the default denoted by [ ]
Please enter a string value for: password ::
Loading Tool: com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server
Login to boot-server as user root was successful.
Download of com.sun.admin.osservmgr.cli.OsServerMgrCli from boot-
server was successful.
Client          Root Area
                Swap Area
                Dump Area
-----
diskless-x86    boot-server:/export/root/diskless-x86
                boot-server:/export/swap/diskless-x86

diskless-sparc  boot-server:/export/root/diskless-sparc
                boot-server:/export/swap/diskless-sparc
```

Note that all directory substructures were formed in the `/export` directory. This is fine, as this directory will not be used for the storage of the diskless clients — a new directory structure will be created using the Scalent import utility.

Configure the Solaris x86 Boot Environment

The `smdiskless add` command used in the previous section does not always correctly configure all subdirectories required for the x86 diskless servers to boot. If needed, the script included in Appendix B, “Config_tftp Script” on page 34 can be used to ensure all files are correctly installed.

Note – This script was included in Vijay S. Upreti's document *Diskless Setup for the Solaris OS for x86 Platforms*, available on the Web at:

http://www.sun.com/bigadmin/features/articles/diskless_x86_setup.pdf

1. Run the `config_tftp` script with the following syntax:

```
config_tftp add diskless-client-name diskless-client-mac-address
```

For example, the following command assumes a diskless client named “diskless-x86” with a MAC address of “00:14:4f:fa:3a:12”:

```
# config_tftp add diskless-x86 00:14:4f:fa:3a:12
```

2. Next, configure DHCP to provide the necessary information that the network-booted clients need to boot. Make sure the DHCP service is enabled and running with no previous settings configured. An easy way to do this is to run the following command that launches the GUI for the DHCP Manager and the wizards that ask for initial configuration information:

```
# /usr/sadm/admin/bin/dhcpmgr
```

Follow the DHCP Manager wizard prompts, as shown in Appendix D, “DHCP Manager Wizard” on page 42, with only the most basic and default information being requested.

Note – Choose ‘no’ when the address range wizard is displayed, as there is no need to select an address range in this case.

- Run the following commands for the initial client, substituting the correct IP address. The `pntadm` command is used to manage the DHCP network tables; the `dhtadm` command is used to manage the DHCP configuration tables.

In this example, the MAC address of the diskless client is `00:14:4f:fa:3a:12`.

```
# pntadm -A diskless-x86 -m 0100144FFA3A12 -f 'MANUAL+PERMANENT'
-i 0100144FFA3A12 IP_ADDRESS

# dhtadm -A -m 0100144FFA3A12
-d ":BootSrvA=boot_server_IP _address:BootFile=0100144FFA3A12:"
```

Initial Boot Configuration of the Diskless Systems

It is recommended to boot the systems at least once, to install the Solaris SPARC and x86 persona software on the disk images to be replicated or moved under the Controller's control. This allows the image to be identified by the Controller when it boots.

- Mount the ISO file corresponding to the SPARC or x86 Solaris Scalent CD on the diskless booted client.
- Run the setup script provided in the Scalent software distribution to install the Scalent Agent software. To do this, copy the 'installation' directory to the filesystem of the diskless client and then run the `setup.sh` command and select the '*P*'ersona option.
- Shut down the servers, as the systems are now configured and ready to be imported.

Configure the Solaris Images on the Controller

On the Solaris NFS server, run the `copysolnb.sh` script in order to prepare the Solaris network-booted persona for the Scalent environment. This script is part of the Solaris CD in the set of Scalent disks.

- Mount the CD and copy the `/installation` directory to a directory in the Solaris NFS server.
- Go to the directory and run the `copysolnb.sh` script as follows — note that the first parameter is the directory where the diskless client was installed previously and the second parameter is the directory that was configured in the Scalent Controller installation.

```
bash-3.2# ./copysolnb.sh /export/root/diskless-x86
172.18.1.100:/shares/nfs/solarisnb/
copying /export/root/diskless-x86 to
172.18.1.100:/shares/nfs/solarisnb/root/diskless-x86
430352 blocks
copying /export/swap/diskless-x86 to
172.18.1.100:/shares/nfs/solarisnb/swap/diskless-x86
copying /export/exec/Solaris_10_i386.all/usr to
172.18.1.100:/shares/nfs/solarisnb/exec/Solaris_10_i386.all/usr
6034880 blocks
```

3. Repeat the process for the SPARC distribution. In this example, a separate folder was created first and then the copy was made to that new folder:

```
bash-3.2# mkdir /shares/nfs/solarisnb/sparc
bash-3.2# ./copysolnb.sh /export/root/diskless-sparc
172.18.1.100:/shares/nfs/solarisnb/sparc
copying /export/root/diskless-sparc to
172.18.1.100:/shares/nfs/solarisnb/sparc/root/diskless-sparc
335552 blocks
copying /export/swap/diskless-sparc to
172.18.1.100:/shares/nfs/solarisnb/sparc/swap/diskless-sparc
copying /export/exec/Solaris_10_sparc.all/usr to
172.18.1.100:/shares/nfs/solarisnb/sparc/exec/Solaris_10_sparc.all/us
r
8173520 blocks
```

4. Next, the Scalent Controller must be made aware of the new images just added. From the /opt/scalent/bin directory, run the SDK, open the configuration and add the persona. Save the configuration before closing the SDK:

```
[root@controller bin]# /opt/scalent/bin/sdk
Setting default port to 8080
>> open
>> add persona bootType=pxegrub_solaris osFamily=solaris osArch=x86_32
image=172.18.1.100:/shares/nfs/solarisnb/root/diskless-x86
swap=172.18.1.100:/shares/nfs/solarisnb/swap/diskless-x86
>> add persona bootType=dhcp_solaris osFamily=solaris osArch=sun4v
image=172.18.1.100:/shares/nfs/solarisnb/sparc/root/diskless-sparc
swap=172.18.1.100:/shares/nfs/solarisnb/sparc/swap/diskless-sparc
>> save
>> exit
[root@controller bin]#
```

5. On the SPARC server that will host the Logical Domains, run the setup.sh script. Choose the option to install the SPARC discovery images and indicate where the NFS server resides and what the installation directory is.
6. Boot both x86 and SPARC clients, then copy the corresponding Scalent .iso and install the persona software. At this point, make these two personas templates and clone them.

Installation for Windows Clients

Servers installed with Windows server software can be booted from the hard drive or from an iSCSI volume available on the network. Regardless of the boot method, the process to get the systems installed starts the same: Systems must be installed on a local disk, followed by the installation of the necessary drivers and the Scalent Persona (agent) software.

For the iSCSI network booted servers, the image must be copied to an iSCSI volume in a similar way as the Linux servers.

Note – At the moment, servers with NVIDIA NICs will not be able to create the necessary VLANs to communicate with the Controller automatically. These NICs can be configured to work correctly by manually enabling ETH802.1P on the advanced properties panel of the NIC. Or, if these types of servers are present in the infrastructure, a PCIe ExpressModule or PCIe card with Intel or Broadcom NICs can be installed to allow proper communication with the Scalent Controller.

Windows Disk-Booted Systems

Installation of a disk-booted system that can be managed by the Controller is very straightforward. The installation is the same as the installation of a normal Windows install, but it is the first step to create the network-booted Windows server.

1. Install the server with Windows 2003 SP1 or R2 and all necessary drivers.
2. Configure one of the NICs to become available on the boot network using a static IP address or DHCP.
3. Next, install the Scalent Persona software:
 - a. Log in as the Administrator, and mount the ISO image through the remote console of the server or physically using the CD drive.
 - b. Copy the installation directory to a location in the hard drive and run the `setup_persona.exe` executable by double clicking it; follow the prompts.
 - c. After the persona installation is completed, the server reboots. Upon boot, the Controller should recognize the disk-booted client/persona.

Windows iSCSI Network-Booted Systems

In order to deploy the Windows Server image on an iSCSI volume, the volume has to be created and installed. This can be done very easily and quickly though the ZFS command line on the storage server that is already running.

The following example assume the ZFS storage pool `shares` has already been created, as shown in “Using Solaris ZFS to provide NFS and iSCSI storage” on page 13.

1. Log in as root user on the ZFS storage node, and run the following `zfs` commands to create the volume:

```
# zfs create -V 10g shares/iscsi-server2
# zfs set shareiscsi=on shares/iscsi-server2
```

2. After the volume is created, confirm that the volume is available to the network and verify the iSCSI target name:

```
bash-3.2# iscsitadm list target
Target: shares/iscsi-server1
    iSCSI Name: iqn.1986-03.com.sun:02:29920fd7-a8f4-6541-c2b8-860a81b31247
    Connections: 1
Target: shares/iscsi-server2
    iSCSI Name: iqn.1986-03.com.sun:02:3dccb172-f061-6525-82c0-a9f40990df6e
    Connections: 0
bash-3.2#
```

In this example two targets are listed: one for the RedHat iSCSI volume created earlier (see “Using Solaris ZFS to provide NFS and iSCSI storage” on page 13), and one for the new volume that will be used for Windows Server 2003. Note that in this example, the RedHat target is in use and it indicates one connection is active to that volume; the new Windows target is not yet active and therefore has zero connections.

3. The next step is to install some tools on Windows that allow the operating system to connect to the target. These tools are included with the Scalent distribution for Windows.
 - a. On the disk-booted Windows server, mount the Scalent CD or use the previously copied 'installation' directory. The Microsoft iSCSI initiator must be installed which will enable the operating system to view the target on the network.
 - b. Change into the installation directory, and run the setup program, `setup_win2003_iscsi_x86.exe`. When installing, make sure the “Initiator Services” and “Software Initiator Options” are checked to be installed. Also, check “Configure iSCSI Network Boot Support” and select the network from which the system should boot.
4. Once the install is finished, configure the initiator by running the configuration utility at Start -> All Programs -> Microsoft iSCSI Initiator -> Microsoft iSCSI Initiator.
 - a. On the Discovery tab of the utility window, click Add under Target Portals and specify the IP address of the iSCSI target server where the volume for the iSCSI-booted Windows persona was created.
 - b. On the Targets tab of the utility window, click Refresh and select the iSCSI target server. Click Log On and OK to log on to the iSCSI target. The Status field should change to Connected. Click OK to accept the changes and close the Microsoft iSCSI Initiator utility.

5. Go to the Disk Management utility by right clicking on Start > My Computer and selecting 'Manage'. The Disk Manager should show the newly added iSCSI target as an available disk drive.
6. In order to convert the image on the hard drive to an iSCSI target bootable device, a utility by emBoot called `winBoot/i` is required. This utility copies the bootable hard drive image to the iSCSI target on the network.
 - a. Open the syscopy utility by choosing Start -> All Programs -> emboot -> winBoot-i Client -> System Copy.
 - b. Choose the default (Disk) mode copy, select the source drive (typically this the drive where the root `C:\` partition is located), select the destination drive and click Proceed.
 - c. Click Yes when the utility prompts you to format the destination drive.
7. At this point, the Windows image is in the iSCSI volume and the disk-booted persona can be deleted on the Controller so that the network booted persona can be added and enabled. To do this, simply click on the disk booted Windows image using the Controller's web based interface and then choose the 'Delete' option from the right pane.
8. Next, add the iSCSI booted persona using the Scalent Controller's SDK using the following syntax:

```
# /opt/scalent/bin/sdk
>> open
>> add persona bootType=iscsi osFamily=windows osArch=x86_32
    image="iscsiwbi|netapp|172.18.1.100|3260|iqn.2003-
    06.com.example:windows.target|iqn.2003-
    06.com.example:windows.initiator|iscsiwbi" kernel="iscsi"
>> save
>> exit
#
```

9. At this point the Windows image is ready to be booted from the iSCSI device. The disk booted system can be rebooted and configured at the BIOS configuration screen to boot from the network instead of the disk drive. The next time the system is booted, it will be discovered as a disk booted machine by the Controller and at this point the Windows iSCSI image/persona can now be assigned to the server.

Summary

This document describes a proof-of-concept exercise setting up a flexible datacenter infrastructure using the Sun Blade 6000 Modular System, Logical Domains, and Scalent V/OE software. The configuration and installation procedures described in this paper are intended as a starting point for readers interested in implementing a dynamic datacenter that includes clients running any combination of Windows, Linux, and Solaris operating systems. Solaris-based applications running on both SPARC and x86 architectures can be easily migrated and deployed across both physical servers as well as virtualized environments such as Logical Domains or VMware.

The addition of Scalent V/OE technology further enhances this environment by enabling blade server modules and rack mount servers to coexist in a dynamic datacenter. The Scalent software also provides a simple and powerful Web GUI interface that organizes all available personas available in a central datacenter library, for easier administration.

These features offer the capability to better manage variable day-to-day workloads as well as making better utilization of data center resources. Servers can be made available, or brought on-line, only when they are providing useful work, and they can be shut down or transitioned to a virtual environment when demand is low, thereby saving on power and cooling expenses.

In today's datacenter, this flexibility and dynamism provides a real advantage when client demand can sky rocket without warning and when workloads can shift from one application to the other. A virtual and dynamic infrastructure, using technologies such as Logical Domain's and the Scalent V/OE software, can provide a means to easily provision and repurpose servers upon demand, to adapt to meet changing demands.

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Appendix A

Supported Ethernet Switches

The following Ethernet switches were supported in Scalent V/OE configurations at the time of publication:

- Foundry Networks BigIron RX Switch, each module with 10/100/1000 ports.
- Foundry Networks FastIron Edge X 424 Switch, with 24 10/100/1000 ports and a variety of options for copper and fibre SFP modules.
- Cisco Catalyst 2950 switches with 12 10/100 ports, such as model: WS-C2950T-12
- Cisco Catalyst 2950 switches with 24 10/100 ports, such as model: WS-C2950-24
- Cisco Catalyst 2950 switches with 24 10/100 ports and 2 100BaseFX ports, such as model: WS-C2950T-24-2FX
- Cisco Catalyst 2950 switches with 24 10/100 ports and 2 10/100/1000 ports, such as model: WS-C2950T-24
- Cisco Catalyst 2950 switches with 24 10/100 ports and 2 SX uplinks: WS-C2950T-24-2SX
- Cisco Catalyst 2950 switches with 48 10/100 ports and 2 10/100/1000 uplinks: WS-C2950T-48-2GE
- Cisco Catalyst 2950 switches with 48 10/100 ports and 2 SX uplinks: WS-C2950T-48-2SX
- Cisco Catalyst 2950G switches with 12 10/100/1000 ports, such as model: WS-C2950G-12
- Cisco Catalyst 2950G switches with 24 10/100/1000 ports, such as model: WS-C2950G-24
- Cisco Catalyst 2950G switches with 24 10/100/1000 ports, using DC power: WS-C2950GDC-24
- Cisco Catalyst 2950G switches with 48 10/100/1000 ports, such as model: WS-C2950G-48
- Cisco Catalyst 2960G switches with 20 10/100/1000 ports and 4 dual-purpose uplinks: WS-C2960G-24TC-L
- Cisco Catalyst 2960G switches with 44 10/100/1000 ports and 4 dual-purpose uplinks: WS-C2960G-48TC-L
- Cisco Catalyst 2970G switches with 24 10/100/1000 ports and 4 dual-purpose uplinks: WS-C2970G-24TS-E
- Cisco Catalyst 3560 switches with 24 10/100 ports and 2 SFP ports, such as model: WS-C3560-24TS-S
- Cisco Catalyst 3560 switches with 24 10/100 PoE ports and 2 SFP ports, such as model: WS-C3560-24PS-S
- Cisco Catalyst 3560 switches with 48 10/100 ports and 4 SFP ports, such as model: WS-C3560-48TS-S
- Cisco Catalyst 3560 switches with 48 10/100 PoE ports and 4 SFP ports, such as model: WS-C3560-48PS-S
- Cisco Catalyst 3560G switches with 24 10/100/1000 ports and 4 SFP ports, such as model: WS-C3560G-24TS-S

- Cisco Catalyst 3560G switches with 24 10/100/1000 PoE ports and 4 SFP ports, such as model: WS-C3560G-24PS-S
- Cisco Catalyst 3560G switches with 48 10/100/1000 ports and 4 SFP ports, such as model: WS-C3560G-48TS
- Cisco Catalyst 3560G switches with 48 10/100/1000 PoE ports and 4 SFP ports, such as model: WS-C3560G-48PS
- Cisco Catalyst 3750 switches with 24 10/100/1000 ports, such as model: WS-C3750G-24T-S
- Cisco Catalyst 3750 switches with 24 10/100/1000 ports and 4 SFP ports, such as model: WS-C3750G-24TS-S
- Cisco Catalyst switches with 3750 48 10/100/1000 ports and 4 SFP ports, such as model: WS-C3750G-48TS-S
- Cisco Catalyst 4948 switches with 48 10/100/1000 ports plus 4 SFP ports, such as model: WS-C4948
- Cisco Catalyst 4948-10G switches with 48 10/100/1000 ports and 2 10-gigabit uplink ports, such as model: WS-C4948-10GE
- Cisco Catalyst 6503 switches, each module with 48 10/100 or 10/100/1000 ports, such as model: WS-C6503
- Cisco Catalyst 6506-E switches, each module with 48 10/100 or 10/100/1000 ports, such as model: WS-C6506-E
- Cisco Catalyst 6504-E switches, each module with 48 10/100 or 10/100/1000 ports, such as model: WS-C6504-E
- Cisco Catalyst 6509-E switches, each module with 48 10/100 or 10/100/1000 ports, such as model: WS-C6509-E
- Cisco Catalyst 6509SP switches, each module with 48 10/100 or 10/100/1000 ports, such as model: WS-C6509SP
- Cisco Catalyst 6509-NEB-A switches, each module with 48 10/100 or 10/100/1000 ports, such as model: WS-C6509-NEBA
- Cisco Catalyst 6509 switches, each module with 48 10/100 or 10/100/1000 ports, such as model: WS-C6509

Appendix B

Config_tftp Script

This script was included in Vijay S. Upreti's document *Diskless Setup for the Solaris OS for x86 Platforms*, available on the Web at:

http://www.sun.com/bigadmin/features/articles/diskless_x86_setup.pdf

Save the following script as `config_tftp` and run with the following syntax:

```
config_tftp add diskless_client_name diskless_client_MAC_address
```

```
#!/usr/bin/sh
#
# Copyright 2005 Sun Microsystems, Inc. All rights reserved.
# Use is subject to license terms.
#
#pragma ident"@(#)config_tftp.sh1.1 05/09/06 SMI"
#
# This script is always invoked by libsmoss with three args:
# <subcmd> <clientname> <client ether address>
# The subcmd must be one of add, delete, and modify.
# The ether address is assumed to be in the usual format of
# x:x:x:x:x:x
# where x is a one or two digit hex number.
#
#
# Convert ETHERADDR to canonical form with hex digits in upper case
#
convert_etheraddr()
{
ether_addr=
for i in 1 2 3 4 5 6; do
hex=`echo ${ETHERADDR} | cut -d : -f $i | tr '[a-f]' '[A-F]`
ether_addr=${ether_addr}:`echo "ibase = 16 ; $hex" | bc`
done
ether_addr=`echo ${ether_addr} | \
    awk -F: '{printf "%02x%02x%02x%02x%02x%02x", $2, $3, $4, $5, $6, $7}`
ETHER_UPPER=`echo ${ether_addr} | tr '[a-f]' '[A-F]`
}
name_to_ipaddr()
{
{
line=`grep -v "^#" /etc/hosts | grep "[ ]$1[ ]`
hostip=`echo $line | (read hostip junk junk junk; echo $hostip)`
if [ X"$hostip" != X ]; then
echo "$hostip"
return 0
fi
echo "ipaddr-for-$1"
return 0
}
}
#
# Modify client's bootenv.rc, build grub menu, create boot archive,
# and lofs mount client's /boot area under /tftpboot/<clientname>.
#
add_client()
{
```

```

# lofs mount /boot of client to /tftpboot
mkdir -p /tftpboot/${CLIENTNAME}
mount -F lofs /export/root/${CLIENTNAME}/boot /tftpboot/${CLIENTNAME}
    echo "/export/root/${CLIENTNAME}/boot - /tftpboot/${CLIENTNAME} lofs
    - yes ro" >>
    /etc/vfstab

#
# setup properties in bootenv.rc
# Note: rootopts should be set to read only to avoid failure of
# SMF boot-archive service.
#
BOOTENVRC=/export/root/${CLIENTNAME}/boot/solaris/bootenv.rc
echo "setprop fstype 'nfsdyn'" >> ${BOOTENVRC}
echo "setprop server-name '${HOSTNAME}'" >> ${BOOTENVRC}
echo "setprop server-path '/export/root/${CLIENTNAME}'" >> ${BOOTENVRC}
echo "setprop server-rootopts 'ro'" >> ${BOOTENVRC}
# create boot archive link to /boot to make them available for tftp
/sbin/bootadm -a update -R /export/root/${CLIENTNAME} 2> /dev/null
rm -f /export/root/${CLIENTNAME}/boot/boot_archive
ln /export/root/${CLIENTNAME}/platform/i86pc/boot_archive \
/export/root/${CLIENTNAME}/boot/boot_archive
if [ ! -f /export/root/${CLIENTNAME}/boot/multiboot ] ; then
ln /export/root/${CLIENTNAME}/platform/i86pc/multiboot \
/export/root/${CLIENTNAME}/boot/multiboot
fi
# setup menu.lst file content
menufile=/export/root/${CLIENTNAME}/boot/grub/menu.lst
rm -f ${menufile}
touch ${menufile}
echo "default=0" >> ${menufile}
echo "timeout=10" >> ${menufile}
echo "title Solaris Diskless Client" >> ${menufile}
echo "root (nd)" >> ${menufile}
echo "# If console is on ttya|ttyb, replace kernel line with" >> ${menufile}
echo "# one of the commented lines" >> ${menufile}
echo "kernel ${CLIENTNAME}/multiboot" >> ${menufile}
echo "# kernel ${CLIENTNAME}/multiboot -B console=ttya" >> ${menufile}
echo "# kernel ${CLIENTNAME}/multiboot -B console=ttyb" >> ${menufile}
echo "module ${CLIENTNAME}/boot_archive" >> ${menufile}
# setup menu.lst.01<ether_addr> link
convert_etheraddr
rm -f /tftpboot/menu.lst.01${ETHER_UPPER}
cp /tftpboot/${CLIENTNAME}/grub/menu.lst /tftp-
boot/menu.lst.01${ETHER_UPPER}
# copy over pxegrub; don't do symlink -- pxegrub must be at top level
rm -f /tftpboot/01${ETHER_UPPER}
cp /tftpboot/${CLIENTNAME}/grub/pxegrub /tftpboot/01${ETHER_UPPER}
echo "\nIf not already configured, enable PXE boot by creating" > /dev/tty
echo "a macro which contains the following values:" > /dev/tty
echo "Boot server IP (BootSrvA) : `name_to_ipaddr ${HOSTNAME}`" > /dev/tty
echo "Boot file      (BootFile) : 01${ETHER_UPPER}" > /dev/tty
echo "\nIf console is on a serial port, edit /tftp-
boot/menu.lst.01${ETHER_UPPER}" > /dev/tty
echo "(see comments in the file)." > /dev/tty
}
#
# The only thing to do is recreate client's boot archive.
#
modify_client()
{

```

```

/sbin/bootadm -a update -R /export/root/${CLIENTNAME}
}
#
# Unmount /tftpboot/<hostname>, delete it from /etc/vfstab, delete
# pxegrub, and menu.lst.
#
delete_client()
{
  umount /tftpboot/${CLIENTNAME}
  rmdir /tftpboot/${CLIENTNAME}
  grep -v "[ ]*/tftpboot/${CLIENTNAME}[ ]*lofs[ ]*" \
    /etc/vfstab > /etc/vfstab.config_tftp
  if [ $? = 0 ] ; then
    mv /etc/vfstab /etc/vfstab-
    mv /etc/vfstab.config_tftp /etc/vfstab
  else
    rm -f /etc/vfstab.config_tftp
  fi
  # The ether addr is not provided on delete, we try to
  # figure it out in order to cleanup.
  ether_upper=`ls -l /tftpboot/menu.lst.* | \
    grep /tftpboot/${CLIENTNAME} | \
    cut -f3 -d. | (read ether junk; echo $ether)`
  if [ X"$ether_upper" != X ] ; then
    rm -f /tftpboot/menu.lst.$ether_upper
    rm -f /tftpboot/$ether_upper
  fi
}
SUBCMD=$1
CLIENTNAME=$2
ETHERADDR=$3
HOSTNAME=`/usr/bin/hostname`
ETHER_UPPER=
# test for pxegrub based boot
if [ ! -f /export/root/${CLIENTNAME}/platform/i86pc/multiboot ] ; then
  exit 0
fi
case $SUBCMD in
  add) add_client ;;
  modify) modify_client ;;
  delete) delete_client ;;
  *) /usr/bin/printf "usage: $0 add|modify|delete <host> <ether>\n"
  exit 1 ;;
esac

```

Appendix C

Scalent V/OE Pre-Installation Checklist

The following tables contain Scalent V/OE pre-installation checklists.

Controller Installation Variables

Table 2. Controller Installation Variables.

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
Variable that tells the installer what kind of rack contains the server where you install the first Controller. Set to any one of the following: • Dell • IBMBladeCenter • Verari • vRack	RACK_TYPE	None; commented out.
A number from 0 to 31 that is unique for this Scalent environment at your location (it must be different from that used for any other Scalent environment that the networks in this environment can connect to). Change to a number from 0 to 31.	SYSTEM_ID	1

Scalent Management Network

The Scalent Controller uses the management network to control chassis management modules and switches, vRack and interconnect switches, server management modules (ALOM, iLO, IPMI, and others), and other elements of the Scalent environment. It's also the network you use from outside the Scalent environment to connect to the Scalent Console.

Table 3. Scalent Management Network.

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
The management network number.	MANAGEMENT_NET	172.16.1.0
The subnet mask of the management network.	MANAGEMENT_NETMASK	255.255.255.0
The virtual IP address on the management network for the Scalent Controller.	VIRTUAL_CONTROLLER_ MANAGEMENT_IP	172.16.1.98

Scalent Control Network

The Scalent Controller and Scalent agent software use this private network to communicate within the Scalent environment. You can't extend this network outside the Scalent environment. However, you must make sure that no other network that can connect to the Scalent environment conflicts with the control network, which uses 10.21.1.0/24. If this network conflicts with another that you can't change, you can specify a different control network when you install Scalent V/OE.

Table 4. Scalent Control Network.

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
The IP address of the Scalent control network. Leave it at the default setting (10.21.1.0 unless you had to change it to avoid a conflict).	CONTROL_NET	10.21.1.0
The broadcast address of the control network.	CONTROL_BROADCAST	10.21.1.255
The subnet mask of the control network.	CONTROL_NETMASK	255.255.255.0

Controller Default Gateway

This setting contains the default gateway address for the controller.

Table 5. Controller Default Gateway.

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
Typically, the IP address of the management network's default gateway. Defaults to the boot network's default gateway if you don't set a value.	CONTROLLER_DEFAULT_GATEWAY	172.16.1.1

Scalent Boot Network

The boot network supports the discovery and network booting of servers in the Scalent environment. The boot network must be separate from and not conflict with any other network used in or reachable by the Scalent environment.

Table 6. Scalent Boot Network.

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
The IP address of the boot network.	BOOT_NET	172.18.1.0
The broadcast address of the boot network.	BOOT_BROADCAST	172.18.1.255
The subnet mask of the boot network.	BOOT_NETMASK	255.255.255.0
The IP address of the boot network's default gateway.	BOOT_GATEWAY	172.18.1.1

*Controller Boot Network Settings**Table 7. Controller Boot Network Settings.*

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
The IP address of the first Controller on the boot network. (You set this when you install and configure the Controller's operating system.)		
The IP address of the second Controller on the boot network. (You set this when you install and configure the Controller's operating system.)		
The start of the range of IP addresses on the boot network for use by the Controller DHCP server.	DHCP_BOOT_IP_RANGE_START	172.16.1.101
The end of the range of IP addresses on the boot network for use by the Controller DHCP server.	DHCP_BOOT_IP_RANGE_END	172.16.1.199

*Management Authentication**Table 8. Management Authentication.*

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
If you are installing on a server in a chassis, the SNMP community string for the Ethernet switch in bay 1. If you are installing on a vRack, the SNMP community string for the first or all vRack switches.	SWITCH_COMMUNITY_STRING	scalent
The telnet username of the chassis or vRack switches.	SWITCH_TELNET_USERNAME	USERID
The telnet password of the chassis or vRack switches.	SWITCH_TELNET_PASSWORD	PASSWORD
The management IP address of the first vRack switch.	SWITCH_MANAGEMENT_IP	172.16.1.78
The standard username that you set when configuring the management modules (ALOM, iLO, IPMI, and so on) in a rack's servers. Leave this variable blank if you are installing the Controller on a Sun V20z or V40z server.	RACK_USER	USERID
The standard password that you set for the management modules when configuring the rack's servers.	RACK_PASSWORD	PASSWORD
The IP address of an (optional) second vRack switch.	SWITCH2_MANAGEMENT_IP	172.16.1.79

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
The SNMP community string for an (optional) second vRack switch.	SWITCH2_COMMUNITY_STRING	scalent
The telnet username for an (optional) second vRack switch. You don't need to set this value if it is the same as for the first vRack switch	SWITCH2_TELNET_USERNAME	USERID
The telnet password for an (optional) second vRack switch. You don't need to set this value if it is the same as for the first vRack switch	SWITCH2_TELNET_PASSWORD	PASSWORD

SPARC Solaris “Discovery” Image Server

If you plan to network-boot Solaris personas on SPARC servers, you need to identify the IP address and path on a network server where you will install the small Solaris image the Controller uses to discover SPARC servers.

Table 9. SPARC Solaris “Discovery” Image Server.

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
The IP address of the server on which you plan to host a SPARC Solaris discovery image, if your deployment plan includes support for network-booted SPARC Solaris personas.	SOLARIS_NAS_IP	172.18.1.13
The path on the server where you plan to host a SPARC Solaris discovery image, if your deployment plan includes support for network-booted SPARC Solaris personas.	SOLARIS_NAS_ROOT	/vol/vol1/solarisnb

VLAN Settings

The default ranges for the VLANs that Scalent manages are correct for most deployments, unless you have a range of VLANs you want to manage yourself; if so, your Scalent representative can help you plan ranges that accommodate unmanaged VLANs.

Table 10. VLAN Settings.

Setting and Notes	Installation Variable Name (If Applicable)	Default Value
The start of the range of VLANs the Scalent Controller uses to create the infrastructure that supports the Scalent environment.	INFRASTRUCTURE_VLAN_START	4002
The end of the range of VLANs the Scalent Controller uses to create the infrastructure that supports the Scalent environment.	INFRASTRUCTURE_VLAN_END	4089
The start of the range of VLANs available to the Scalent Controller to create components of virtual networks.	SCALENT_ASSIGNED_VLAN_START	3746
The end of the range of VLANs available to the Scalent Controller to create components of virtual networks.	SCALENT_ASSIGNED_VLAN_END	4001
The start of the range of VLANs that you can assign to virtual networks in the Scalent environment.	CUSTOMER_ASSIGNED_VLAN_START	2
The end of the range of VLANs that you can assign to virtual networks in the Scalent environment.	CUSTOMER_ASSIGNED_VLAN_END	3745
The VLAN of the Scalent control network.	CONTROL_VLAN	4002
The VLAN of the Scalent management network.	MANAGEMENT_VLAN	4003
The VLAN of the Scalent boot network.	BOOT_VLAN	4004
The VLAN of the Scalent VLAN sink, the VLAN for unused ports.	SVS_VLAN	4005

vWWPN Creation

The Controller uses an algorithm based on the Scalent organizationally unique ID and the system ID to create unique virtual world-wide port names (vWWPNs) for retargetable SAN-booted personas and VMRacks in your Scalent environment. In rare cases, your Scalent representative may advise you to override that algorithm and set a different prefix for creating vWWPNs.

Table 11. vWWPN Creation.

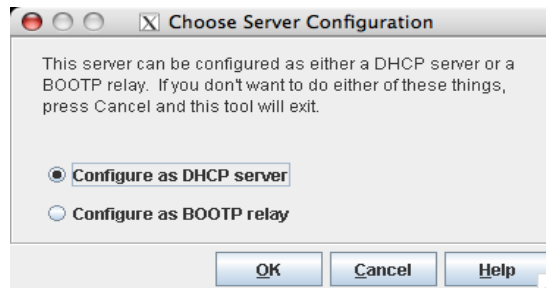
Setting and Notes	Installation Variable Name (If Applicable)	Default Value
Do not set this value without assistance from your Scalent representative.	VWWPN_PREFIX	Not set; commented out.

Appendix D

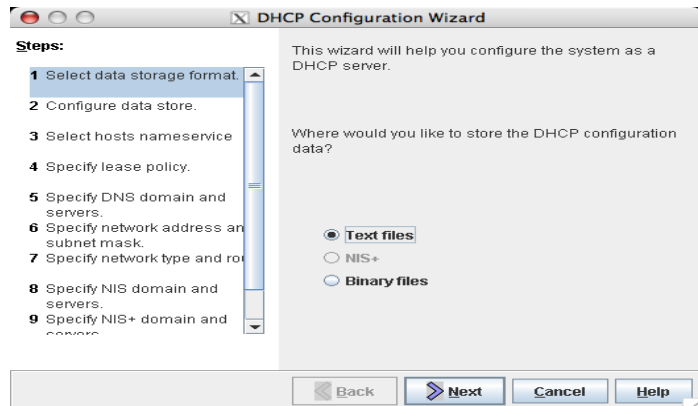
DHCP Manager Wizard

The following screen shots illustrate an example session of running the DHCP Configuration Wizard:

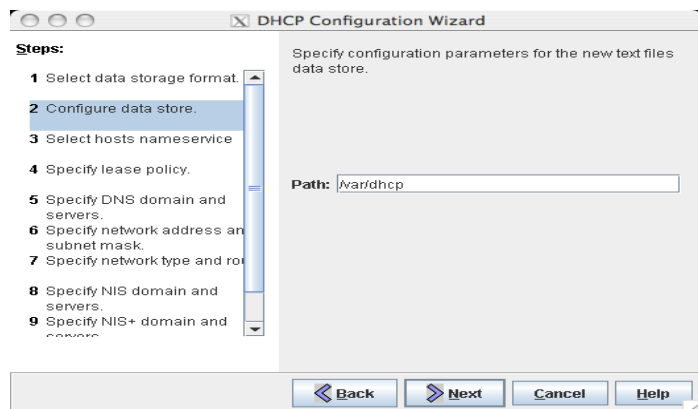
Screen 1:



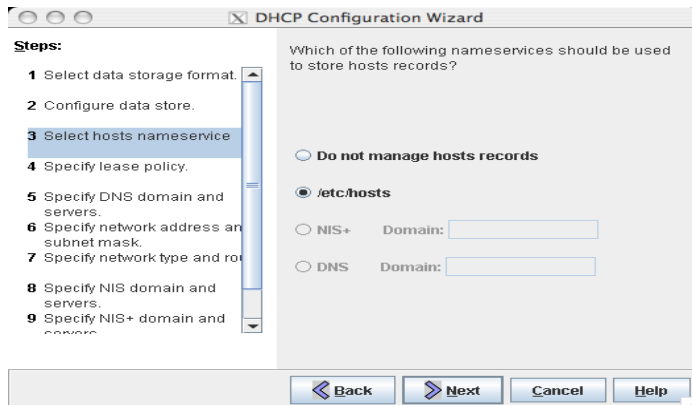
Screen 2:



Screen 3:



Screen 4:



Steps:

- 1 Select data storage format.
- 2 Configure data store.
- 3 Select hosts nameservice**
- 4 Specify lease policy.
- 5 Specify DNS domain and servers.
- 6 Specify network address and subnet mask.
- 7 Specify network type and router.
- 8 Specify NIS domain and servers.
- 9 Specify NIS+ domain and servers.

Which of the following nameservices should be used to store hosts records?

☐ Do not manage hosts records

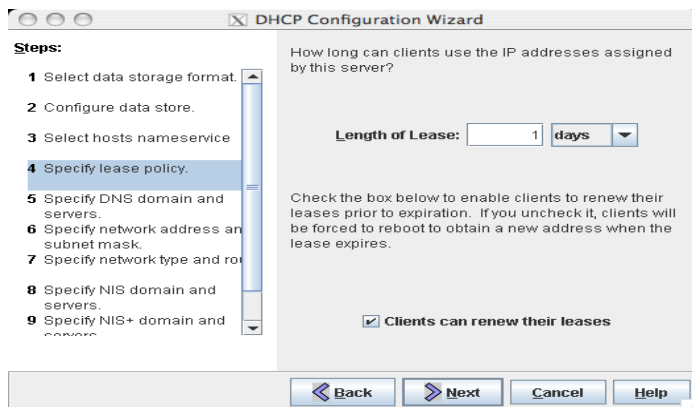
☒ /etc/hosts

☐ NIS+ Domain:

☐ DNS Domain:

Back Next Cancel Help

Screen 5:



Steps:

- 1 Select data storage format.
- 2 Configure data store.
- 3 Select hosts nameservice
- 4 Specify lease policy.**
- 5 Specify DNS domain and servers.
- 6 Specify network address and subnet mask.
- 7 Specify network type and router.
- 8 Specify NIS domain and servers.
- 9 Specify NIS+ domain and servers.

How long can clients use the IP addresses assigned by this server?

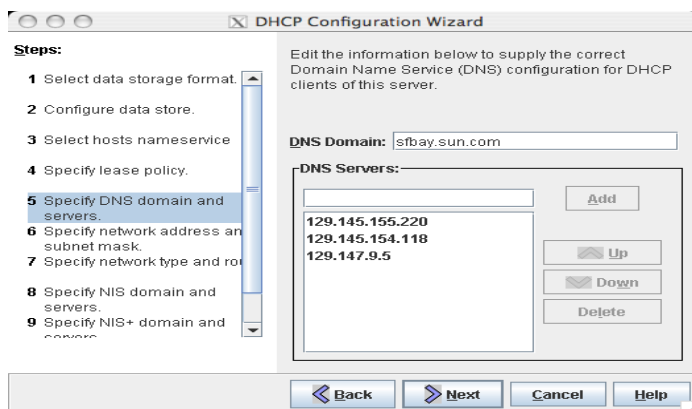
Length of Lease: days

Check the box below to enable clients to renew their leases prior to expiration. If you uncheck it, clients will be forced to reboot to obtain a new address when the lease expires.

☒ Clients can renew their leases

Back Next Cancel Help

Screen 6:



Steps:

- 1 Select data storage format.
- 2 Configure data store.
- 3 Select hosts nameservice
- 4 Specify lease policy.
- 5 Specify DNS domain and servers.**
- 6 Specify network address and subnet mask.
- 7 Specify network type and router.
- 8 Specify NIS domain and servers.
- 9 Specify NIS+ domain and servers.

Edit the information below to supply the correct Domain Name Service (DNS) configuration for DHCP clients of this server.

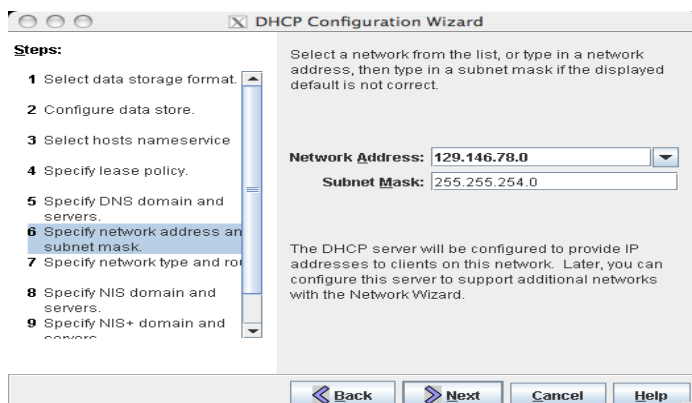
DNS Domain:

DNS Servers:

129.145.155.220	Add Up Down Delete
129.145.154.118	
129.147.9.5	

Back Next Cancel Help

Screen 7:



Steps:

- 1 Select data storage format.
- 2 Configure data store.
- 3 Select hosts nameservice
- 4 Specify lease policy.
- 5 Specify DNS domain and servers.
- 6 Specify network address and subnet mask.**
- 7 Specify network type and router.
- 8 Specify NIS domain and servers.
- 9 Specify NIS+ domain and servers.

Select a network from the list, or type in a network address, then type in a subnet mask if the displayed default is not correct.

Network Address:

Subnet Mask:

The DHCP server will be configured to provide IP addresses to clients on this network. Later, you can configure this server to support additional networks with the Network Wizard.

Back Next Cancel Help

Screen 8:

Steps:

- 1 Select data storage format.
- 2 Configure data store.
- 3 Select hosts nameservice.
- 4 Specify lease policy.
- 5 Specify DNS domain and servers.
- 6 Specify network address and subnet mask.
- 7 Specify network type and routing.**
- 8 Specify NIS domain and servers.
- 9 Specify NIS+ domain and servers.

Please enter the following information for the network:

Network Type

☒ Local-Area (LAN)

☐ Point-to-Point

Routing

☒ Use router discovery protocol

☐ Use router:

Navigation: Back, Next, Cancel, Help

Screen 9:

Steps:

- 1 Select data storage format.
- 2 Configure data store.
- 3 Select hosts nameservice.
- 4 Specify lease policy.
- 5 Specify DNS domain and servers.
- 6 Specify network address and subnet mask.
- 7 Specify network type and routing.
- 8 Specify NIS domain and servers.**
- 9 Specify NIS+ domain and servers.

Edit the information below to supply the correct Network Information Service (NIS) configuration to DHCP clients on this network.

NIS Domain:

NIS Servers:

Navigation: Back, Next, Cancel, Help

Screen 10:

Steps:

- 1 Select data storage format.
- 2 Configure data store.
- 3 Select hosts nameservice.
- 4 Specify lease policy.
- 5 Specify DNS domain and servers.
- 6 Specify network address and subnet mask.
- 7 Specify network type and routing.
- 8 Specify NIS domain and servers.
- 9 Specify NIS+ domain and servers.**

Edit the information below to supply the correct Network Information Service Plus (NIS+) configuration to DHCP clients on this network.

NIS+ Domain:

NIS+ Servers:

Navigation: Back, Next, Cancel, Help

Screen 11:

Steps:

- 2 Configure data store.
- 3 Select hosts nameservice.
- 4 Specify lease policy.
- 5 Specify DNS domain and servers.
- 6 Specify network address and subnet mask.
- 7 Specify network type and routing.
- 8 Specify NIS domain and servers.
- 9 Specify NIS+ domain and servers.
- 10 Review.**

The system will be configured as a DHCP server with the following settings:

Data Storage: Text files

Hosts Nameservice: /etc/hosts

Length of Lease: 1 days, renewable

DNS Domain: sfbay.sun.com

DNS Servers: 129.145.155.220, 129.145.154.1

Network Address: 129.146.78.0

Subnet Mask: 255.255.254.0

Network Type: Local-Area (LAN)

Router: Use router discovery protocol

NIS Domain:

NIS Servers:

NIS+ Domain:

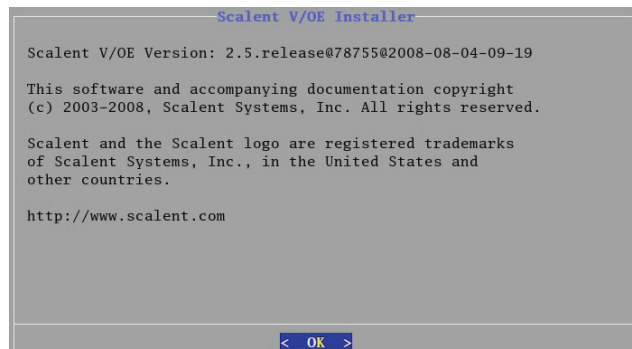
NIS+ Servers:

Appendix E

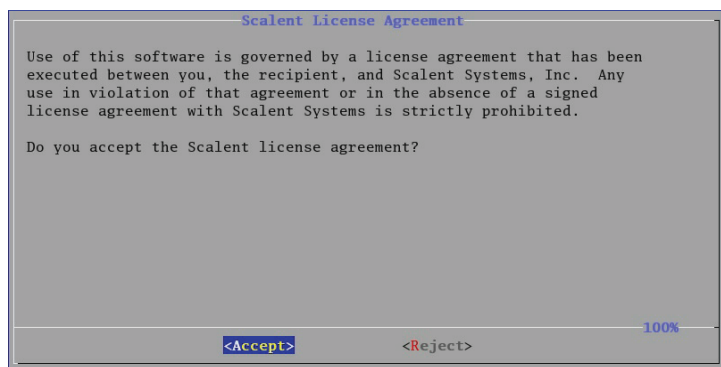
Scalent Controller Installation

The following screen shots illustrate an example session of running the Scalent Installation GUI:

Screen 1:



Screen 2:



Screen 3:



Screen 4:

Please select the initial rack type for the install:

Dell	Dell Chassis
IBM BladeCenter	IBM BladeCenter Chassis
vRack	Virtual Rack

< **K** > < Exit >

Screen 5:

INFO

Will you be configuring the second rack switch now?

< Yes > < **No** >

Screen 6:

INFO

Will you be network booting Solaris Sparc?

< **Yes** > < No >

Screen 7:

INFO

Will you be customizing the VLANS that the Controller uses?

< Yes > < **No** >

Screen 8:

"Please fill in the form in its entirety.
Use <CTRL>+<N> for next field, <CTRL>+<P> for previous."

SYSTEM_ID	1
MANAGEMENT_NET	10.6.160.0
MANAGEMENT_NETMASK	255.255.252.0
VIRTUAL_CONTROLLER_MANAGEMENT_IP	10.6.160.98
CONTROLLER_DEFAULT_GATEWAY	10.6.163.254
SWITCH_MANAGEMENT_IP	10.6.160.78
SWITCH_COMMUNITY_STRING	scalent
SWITCH_TELNET_USERNAME	scalent
SWITCH_TELNET_PASSWORD	scalent
RACK_USER	root
RACK_PASSWORD	changeme
DNS_ADDRS	129.145.154.118 129.

1(+)

< Next > < Cancel > < Help >

Screen 9:

"Please fill in the form in its entirety.
Use <CTRL>+<N> for next field, <CTRL>+<P> for previous."

CONTROL_NET	10.21.1.0
CONTROL_BROADCAST	10.21.1.255
CONTROL_NETMASK	255.255.255.0

< Next > < Prev > < Cancel > < Help >

Screen 10:

"Please fill in the form in its entirety.
Use <CTRL>+<N> for next field, <CTRL>+<P> for previous."

BOOT_NET	172.18.1.0
BOOT_BROADCAST	172.18.1.255
BOOT_NETMASK	255.255.255.0
BOOT_GATEWAY	172.18.1.1
DHCP_BOOT_IP_RANGE_START	172.18.1.101
DHCP_BOOT_IP_RANGE_END	172.18.1.199

< Next > < Prev > < Cancel > < Help >

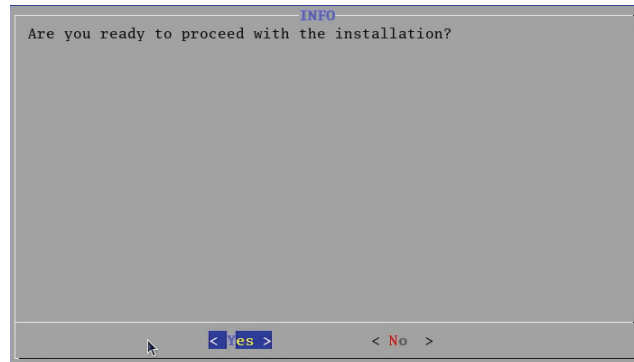
Screen 11:

"Please fill in the form in its entirety.
Use <CTRL>+<N> for next field, <CTRL>+<P> for previous."

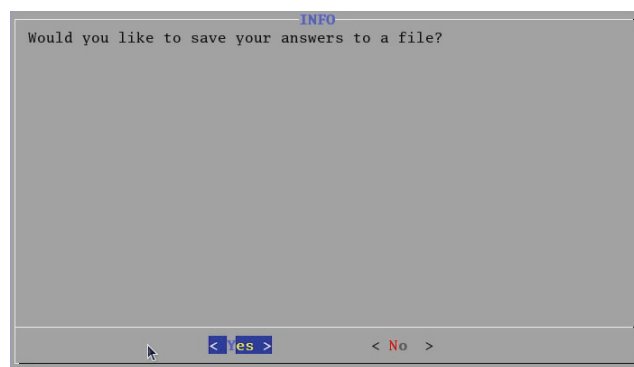
SOLARIS_NAS_IP	172.18.1.100
SOLARIS_NAS_ROOT	/shares/nfs/solarisn

< Next > < Prev > < Cancel > < Help >

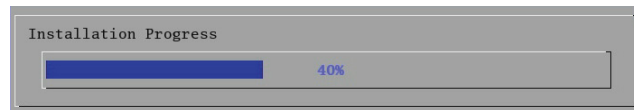
Screen 12:



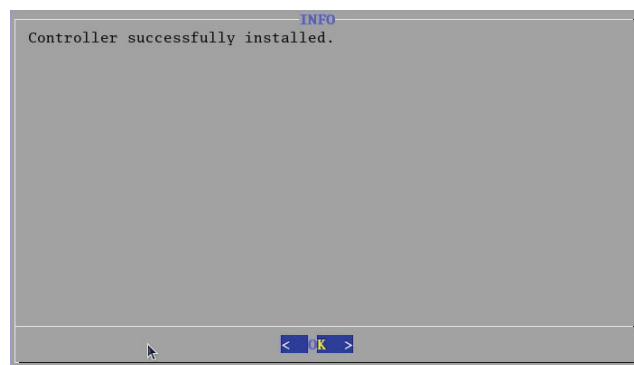
Screen 13:



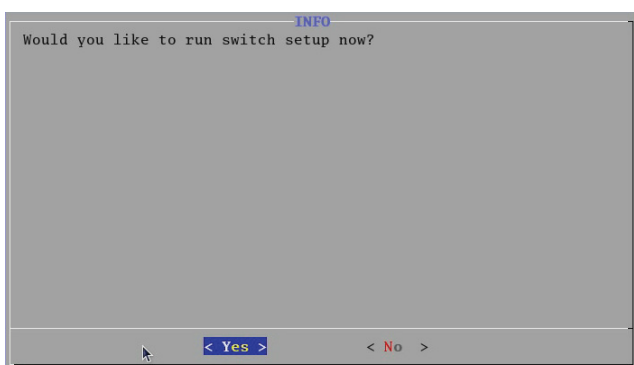
Screen 14:



Screen 15:



Screen 16:



Screen 17:



Note – This screen will only appear if there is a problem and the switch setup fails. If this occurs, follow the displayed directions to run the `doSwitchSetup` script manually. The alternate switch configuration (below) will be used.

```
Switch 10.6.160.78 Model: Cisco WS-C3750G-48TS-S
Port Roles for Module 1:

Gi1/0/1 management  Gi1/0/19 server    Gi1/0/37 server
Gi1/0/2 server       Gi1/0/20 server    Gi1/0/38 server
Gi1/0/3 server       Gi1/0/21 server    Gi1/0/39 server
Gi1/0/4 server       Gi1/0/22 server    Gi1/0/40 server
Gi1/0/5 server       Gi1/0/23 server    Gi1/0/41 external
Gi1/0/6 server       Gi1/0/24 server    Gi1/0/42 external
Gi1/0/7 server       Gi1/0/25 server    Gi1/0/43 external
Gi1/0/8 server       Gi1/0/26 server    Gi1/0/44 external
Gi1/0/9 server       Gi1/0/27 server    Gi1/0/45 external
Gi1/0/10 server      Gi1/0/28 server    Gi1/0/46 external
Gi1/0/11 server      Gi1/0/29 server    Gi1/0/47 external
Gi1/0/12 server      Gi1/0/30 server    Gi1/0/48 external
Gi1/0/13 server      Gi1/0/31 server    Gi1/0/49 external
Gi1/0/14 server      Gi1/0/32 server    Gi1/0/50 external
Gi1/0/15 server      Gi1/0/33 server    Gi1/0/51 external
Gi1/0/16 server      Gi1/0/34 server    Gi1/0/52 external
Gi1/0/17 server      Gi1/0/35 server
Gi1/0/18 server      Gi1/0/36 server

s = server, e = external, u = unmanaged, m = management
Press Enter when done
```

Screen 18:



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